The showhanmengrid package

Show the hanmen grid of Japanese text and its overflow on every page

Munehiro Yamamoto
https://github.com/munepi/showhanmengrid

v1.0 (2026/10/25)

1 Introduction

The *hanmen* (kihon hanmen, 基本版面), the basic text block of Japanese typesetting, is built from four values: the character size (1 zw, the width of a full-width character), the number of characters per line, the number of lines per page and the line feed. When

- `\textwidth` is a multiple of 1 zw, and
- `\textheight` is `\topskip` plus a multiple of `\baselineskip`,

every character of the body text sits on a fixed grid: the text is *set solid* (*beta-gumi*, ベタ組み).

The showhanmengrid package draws, in the background of every page,

1. in cyan, one square of 1 zw × 1 zw for each character position, and the frame of the whole grid;
2. in magenta, the *overflow*: the part of the text area that does not fit the grid.

Both horizontal (yoko, 横組み) and vertical (tate, 縦組み) writing are supported.

Supported formats/engines pLaTeX and upLaTeX (with dvipdfmx), and LuaLaTeX with LuaTeX-ja. LaTeX2e 2025-06-01 or later is required. With any other engine (pdfLaTeX, XeLaTeX, or LuaLaTeX without LuaTeX-ja) the package issues a warning and does nothing.

2 Usage

```
\usepackage{showhanmengrid}
```

Nothing else is needed: the grid is computed at `\begin{document}` from the layout parameters of the class, and the writing direction is taken from the document (`\tate` of pTeX, `direction=3` of LuaTeX-ja). Keys can be given as package options or with `\showhanmengridsetup` in the preamble:

```
\usepackage[color=blue!50, overflow=false]{showhanmengrid}
\showhanmengridsetup{rulewidth=0.2pt}
```

Key	Default	Description
`color`	`cyan`	color of the grid
`overflowcolor`	`magenta`	color of the overflow
`rulewidth`	`0.1pt`	width of the grid lines
`frame`	`true`	draw the frame of the whole grid
`overflow`	`true`	draw the overflow
`chars`	(computed)	characters per line
`lines`	(computed)	lines per page
`ascent`	`0.88zw`	yoko: height of the character face above the baseline

3 How the grid is computed

Let 1 zw and b be the full-width character size and `\baselineskip` of `\normalsize`. In both yoko and tate, `\textwidth` is the length of a line and `\textheight` the extent over which the lines are stacked, so that

$$\begin{aligned} \text{chars} &= \lfloor \texttt{\textwidth}/1\,\text{zw} \rfloor, \\ \text{lines} &= \lfloor (\texttt{\textheight} - \texttt{\topskip})/b \rfloor + 1, \end{aligned}$$

and the grid is chars × 1 zw long along the lines and (lines − 1) b + 1 zw wide across the lines. A remainder of less than 0.01 pt is ignored (jlreq, for instance, adds 0.0001 pt to `\textwidth`). Both values and the overflow are written to the log.

Yoko The grid begins at the top of the character face of the first line: `\topskip` minus the ascent (0.88 zw by default) below the top of the text area. The lines run from top to bottom.

Tate The baseline of the first line is `\topskip` left of the right edge of the text area, and the character face extends 0.5 zw on each side of the baseline. The lines run from right to left.

Side margins The text area of the current page is placed with `\oddsidemargin` or `\evensidemargin` as the LaTeX kernel does it. The `\@outputpage` of pLaTeX has a branch that would swap the two margins in tate, but it tests `\iftdir` inside `\vbox{\yoko ...}` and so never applies; the package follows what is actually typeset.

4 The overflow

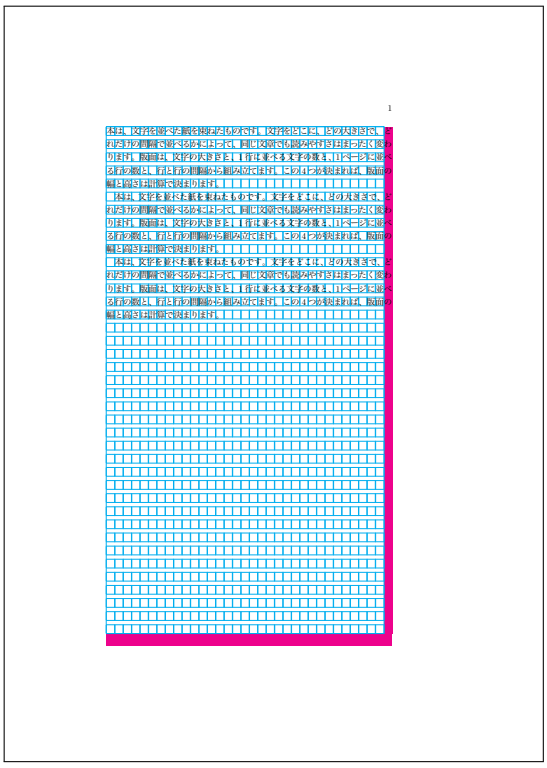
Along the lines `\textwidth` − chars × 1 zw, drawn after the end of the lines (right in yoko, bottom in tate). It is distributed into the inter-character spaces of every justified line: the characters drift away from their squares.

Across the lines `\textheight` − `\topskip` − (lines − 1) b, drawn after the last line (bottom in yoko, left in tate). The page is not filled by whole lines; with `\flushbottom` the remainder is distributed into vertical glue.

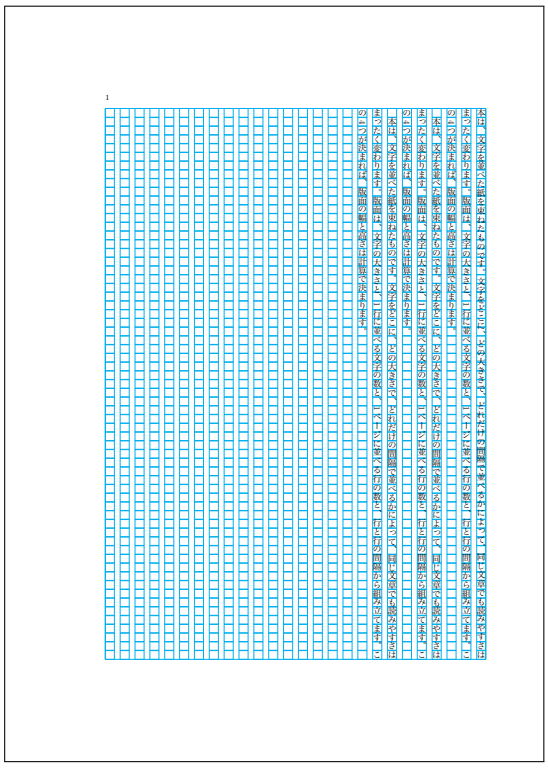
A text area with an overflow is what you get when only the four margins are given, as in word processors or with the `geometry` package. The default layout of the classic Japanese classes is another example: jbook (pLaTeX), ujbook (upLaTeX) and ltjbook (LuaTeX-ja) take their `\textwidth` from the standard book class, min(`\paperwidth` − 2 in, 327 pt), a value chosen for Computer Modern; at 10 pt it is 33.98 zw, i.e. 9.47 pt longer than 33 characters. Their `\textheight`, $\lfloor (\texttt{\paperheight} - 3.5\,\text{in})/b \rfloor\, b +$ `\topskip`, is then truncated to whole points by `\@settopoint`; since `\topskip` is not a whole number of points, the truncation makes the last line no longer fit, and one line is lost (39 lines instead of 40 on A4, with an overflow of about 14 pt). Only the `compatibility`-mode sizes of these classes use multiples of `\Cwd`. jsbook and ltjsbook set `\textwidth` to a multiple of 1 zw, and jlreq sets the text solid by itself.

5 Limitations

- Two-column layout is not supported: the grid spans `\textwidth`.
- Trim marks that shift the page (`tombo` of jclasses, jlreq-trimmarks) are not taken into account.
- The grid is computed once, at `\begin{document}`; changes of `\textwidth` or `\textheight` later in the document are ignored.



ujbook, default layout (A4): 33 chars × 39 lines with an overflow in both directions



jlreq, `tate`: 63 chars × 26 lines, set solid

6 Tests

`make test` compiles every test file with pLaTeX, upLaTeX (+ dvipdfmx) and LuaLaTeX and checks the log. `SHG-EXPECT OK` means that the writing direction, the number of characters and lines, and the overflow are as expected; `SHG-POS OK` means that the first character of the page, measured with `\pdfsavepos` (pTeX) or `\savepos` (LuaTeX), sits in the top left square of the grid as drawn, on odd and even pages. With LuaTeX-ja in tate this check is skipped (`SHG-POS SKIP`): LuaTeX-ja typesets tate text in a rotated box whose `\savepos` coordinates are not those of the page.

7 History

v1.0 (2026/10/25) First public release. Generalized from the version written for the book on LaTeX page layout: it no longer needs `\hanmenwidth`/`\hanmenheight`, supports tate and pLaTeX, and draws the overflow.

8 License

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